

EXP. NO. 3

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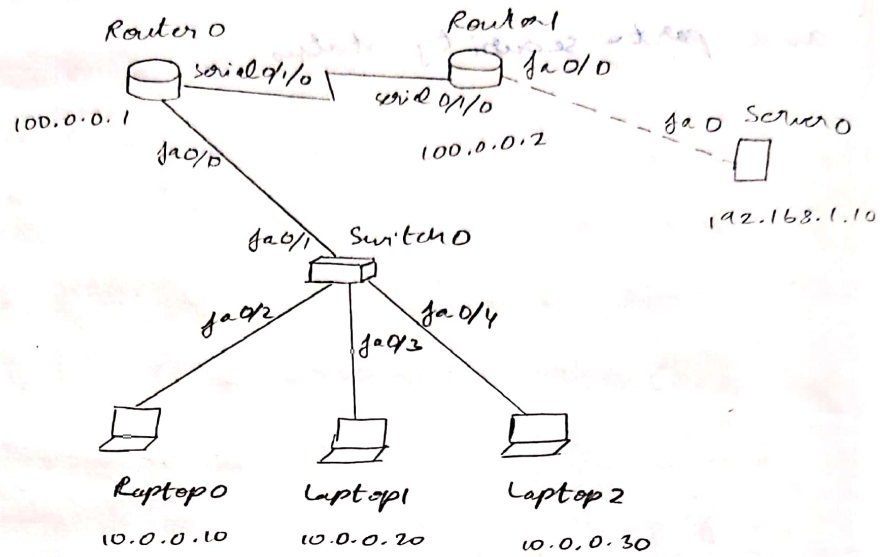
CONFIGURE STATIC NAT IN CISCO PACKET TRACER

AIM

To configure and verify static NAT (Network Address Translation) in Cisco Packet Tracer and understand the NAT address mapping between inside local, inside global, outside local and outside global addresses.

PROCEDURE

1. Create a Network Topology with three laptops, two routers (R1 and R2), and a server as per the given IP addressing table.
2. Set static IP addresses on laptops (inside local), Set IP address on server (outside local) and configure interfaces on both routers.
3. Enable both Fast Ethernet and serial interfaces on the routers.
4. Configure DCE end for serial links to apply clock rate if needed.
5. Map laptop IP (inside local) to public IP (inside global) on R1. Define Fast Ethernet as the inside interface and serial interface as the outside interface.
6. Map server IP (inside local) to global IP (outside global). Define Fast Ethernet as inside, serial as outside.

TOPOLOGY

OUTPUT

(i) From Laptop 0:-

~~C:\>ping 192.~~

C:\>ping 200.0.0.10

Pinging 200.0.0.10 with 32 bytes of data:

Reply from 200.0.0.10: bytes = 32 time = 1ms TTL = 126

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Reply from 200.0.0.10: bytes = 32 time = 1ms TTL = 126

Ping statistics for 200.0.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

minimum = 1ms, maximum = 1ms, Average = 1ms

C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 10.0.0.1: Destination host unreachable.

Reply from 10.0.0.1: Destination host unreachable.

Reply from 10.0.0.1: Destination host unreachable.

Request timed out.

Ping statistics for 192.168.1.10:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

(ii) Router 1:-

R1# show ip nat translations

	Pro	Inside global	Inside local	Outside local	Outside global
icmp		50.0.0.10	10.0.0.10	200.0.0.10	200.0.0.10

(iii) From Laptop 1:-

Ping 200.0.0.10 → Fail

7. On R1, route to Server's network via R2 and on R2, route to Laptop's public IP via R1.
8. From Laptop 0, ping the Server's global IP. Try accessing the Server via web browser using the global IP. Repeat the same with Laptop 1 and Laptop 2.
9. Check translation table on routers and observe how NAT hides actual IPs behind mapped global IPs.

RESULT

Static NAT was successfully configured and verified. Only the mapped internal host was able to communicate with the external server using global IP. The NAT translation table correctly showed static IP mappings. This best demonstrates how NAT enables selective exposure of internal IPs to external networks.